

A proof of Conjecture 00000006630

Two hard-core lattice gases with identical correlation inequalities but different equilibria

1 The conjecture

Conjecture 00000006630 reads:

Definition: Correlation inequalities and equilibria are two layers. *Conjecture:* There exist two lattice-gas models with identical inequalities but different equilibria, and the separation is realized by an explicit violation with the same inequalities but different configuration counting.

It is an existential statement. We prove it with two explicit hard-core lattice gases on two adjacent sites.

2 Reading and definitions

Definition 1 (Hard-core lattice gas). A lattice-gas model on a finite site set V is a simple graph G on V (the exclusion graph) together with an activity $\lambda > 0$. A configuration is the set $\eta \subseteq V$ of occupied sites; it is *allowed* if no two occupied sites are adjacent in G . The weight of η is $\lambda^{|\eta|}$ if η is allowed and 0 otherwise.

Definition 2 (Configuration count, equilibrium). The *configuration count* (partition function) is $Z = \sum_{\eta} \text{weight}(\eta)$. The *equilibrium* is the Gibbs state $\mu(\eta) = \text{weight}(\eta)/Z$, and $\langle f \rangle = \sum_{\eta} f(\eta)\mu(\eta)$.

Definition 3 (Correlation inequalities). For $B \subseteq V$ let $n_B(\eta) = [B \subseteq \eta]$ be the occupation monomial. The *correlation-inequality profile* of the model is

$$\text{profile}(B, C) = \text{sign}(\langle n_B n_C \rangle - \langle n_B \rangle \langle n_C \rangle) \in \{-1, 0, 1\} \quad (B, C \subseteq V).$$

It records, for every pair of occupation monomials, which of $>$, $=$, $<$ holds between $\langle n_B n_C \rangle$ and $\langle n_B \rangle \langle n_C \rangle$; these are the FKG / Harris / Griffiths-type correlation inequalities. Two models have *identical inequalities* if their profiles are equal as functions.

3 Results

Lemma 4. With $W(B) = \sum_{\eta} n_B(\eta) \text{weight}(\eta)$, we have $Z > 0$ and $\text{profile}(B, C) = \text{sign}(ZW(B \cup C) - W(B)W(C))$.

Proof. $Z \geq \text{weight}(\emptyset) = 1$. Since $n_B n_C = n_{B \cup C}$, we get $\langle n_B n_C \rangle - \langle n_B \rangle \langle n_C \rangle = (ZW(B \cup C) - W(B)W(C))/Z^2$, and dividing by $Z^2 > 0$ keeps the sign. $\square$

Let $K_2(\lambda)$ be the model on $V = \{0, 1\}$ with the two sites adjacent and activity λ . Its allowed configurations are $\emptyset, \{0\}, \{1\}$, so $Z = 1 + 2\lambda$ and $W(B) = n_B(\emptyset) + \lambda n_B(\{0\}) + \lambda n_B(\{1\})$.

Lemma 5 (Profile is independent of the activity). *For every $\lambda > 0$ and $B, C \subseteq \{0, 1\}$: $\text{profile}(B, C) = 0$ if one of B, C is $\emptyset$ or $\{0, 1\}$; otherwise it is 1 if $B = C$ and -1 if $B \neq C$. Hence all models $K_2(\lambda)$, $\lambda > 0$, have identical correlation inequalities.*

Proof. By Lemma 4 we check the 16 pairs. If $B = \emptyset$, $ZW(C) - ZW(C) = 0$. If $B = \{0, 1\}$, $W(B) = W(B \cup C) = 0$. If $B = C = \{0\}$, $(1 + 2\lambda)\lambda - \lambda^2 = \lambda + \lambda^2 > 0$. If $B = \{0\}$, $C = \{1\}$, $B \cup C = \{0, 1\}$ and the value is $-\lambda^2 < 0$. The remaining cases are symmetric. $\square$

Theorem 6 (Conjecture 00000006630). *The models $M_1 = K_2(1)$ and $M_2 = K_2(2)$ satisfy:*

1. *identical inequalities:* $\text{profile}_{M_1} = \text{profile}_{M_2}$;
2. *different equilibria:* $\mu_{M_1}(\{0\}) = 1/3 \neq 2/5 = \mu_{M_2}(\{0\})$;
3. *the same explicit violation in both:* $\langle n_0 n_1 \rangle < \langle n_0 \rangle \langle n_1 \rangle$, i.e. the positive-correlation (FKG/Harris) inequality fails at the adjacent pair;
4. *different configuration counts:* $Z_{M_1} = 3$ and $Z_{M_2} = 5$.

Proof. (1) and (3) follow from Lemma 5 (the value -1 at $(\{0\}, \{1\})$). (2): $\mu(\{0\}) = \lambda/(1 + 2\lambda)$. (4): $Z = 1 + 2\lambda$. $\square$

Remark 7 (Literal counts). For an integer activity q , $Z = 1 + 2q$ is the number of configurations of the two-site exclusion gas whose particles carry one of q internal states. In Lean, the type of maps $\omega : \{0, 1\} \rightarrow \{\text{empty}\} \cup \{1, \dots, q\}$ with at most one occupied site has cardinality 3 for $q = 1$ and 5 for $q = 2$ (`card_internal_one`, `card_internal_two`). Moreover, if $N_q(\eta)$ is the number of such configurations whose set of occupied sites is η , then $\mu_{K_2(q)}(\eta) = N_q(\eta)/(1 + 2q)$ for every η : the equilibrium is the occupation marginal of the uniform distribution on these configurations (`gibbs_eq_count_one`, `gibbs_eq_count_two`; the counts are $N_q = 1, q, q, 0$ for $\eta = \emptyset, \{0\}, \{1\}, \{0, 1\}$, and the Lean name of N_q is `countOcc`).

4 Lean formalization

The file `lean/Conjecture6630/Basic.lean` (Lean 4, Mathlib v4.33.1, namespace `Submission00000006630`) defines `LatticeGas`, `Allowed`, `weight`, `Z`, `gibbs`, `expect`, `occ`, `cov`, `profile` (a `SignType`-valued function) and `W`, exactly as above. The main statements are `profile_eq` (Lemma 4), `profile_K2` and `profile_K2_eq` (Lemma 5), `violation_K2`, and

`theorem conjecture6630 :`

```
exists M1 M2 : LatticeGas (Fin 2),
  profile M1 = profile M2 /\
  gibbs M1 != gibbs M2 /\
  (expect M1 (fun e => occ {0} e * occ {1} e)
    < expect M1 (occ {0}) * expect M1 (occ {1})) /\
  (expect M2 (fun e => occ {0} e * occ {1} e)
    < expect M2 (occ {0}) * expect M2 (occ {1})) /\
  Z M1 = 3 /\ Z M2 = 5 /\
  Fintype.card (InternalConfig 1) = 3 /\
  Fintype.card (InternalConfig 2) = 5 /\
  (forall e, gibbs M1 e
    = countOcc 1 e / Fintype.card (InternalConfig 1)) /\
```

```
(forall e, gibbs M2 e
  = countOcc 2 e / Fintype.card (InternalConfig 2))
```

(written here in ASCII; the Lean source uses Unicode symbols).

Axiom audit. lake build succeeds. #print axioms for conjecture6630, profile_K2_eq and violation_K2 reports only propext, Classical.choice and Quot.sound. There is no sorry, no native_decide and no added axiom.

5 Scope

“Lattice-gas model” is read as a hard-core (exclusion) lattice gas in the grand-canonical ensemble, “equilibrium” as its Gibbs state, “correlation inequalities” as the full sign profile of truncated correlations of occupation monomials, “explicit violation” as a violation of the positive-correlation (FKG/Harris) inequality shared by both models, and “configuration counting” as the partition function, which for integer activities is a literal count. The two models have the same set of allowed occupation patterns (three); they differ in the weighted count, which for $q = 1, 2$ is the literal number of internal-state configurations. Readings with soft (finite-energy) interactions or with infinite-volume Gibbs measures are not treated.